STAT5291 Advanced Data Analysis

Final Project

Project Teammates

Yanrui Wang (yw4543)
Yici Chen (yc4818)
Yujia Liu (yl5966)
Xiaojing Yang (xy2714)
Xinyi Shi (xs2594)

1. Introduction

Customer segmentation is important for businesses because different individuals exhibit different shopping behaviors. In this project, unsupervised learning methods (clustering) are used to group mall customers based on age, annual income, and spending score. Instead of treating all customers as a single group, these segments provide insights into how to design more effective and targeted marketing strategies.

Exploratory analysis shows that annual income and spending score display clear patterns(Figure1). For example, customers with similar income levels may have very different spending behaviors, with some high-income individuals spending significantly more than others(Figure2). To identify these underlying structures, four clustering methods, including K-Means, Gaussian Mixture Model, Spectral Clustering and Hierarchical Clustering are compared to evaluate which approach best captures customer behavior.

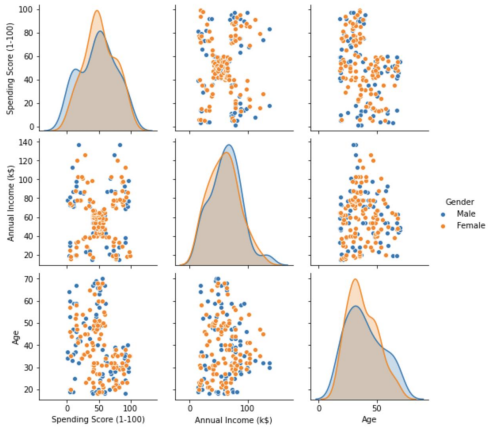


Figure1 Pair plot of Age, Annual Income, and Spending Score

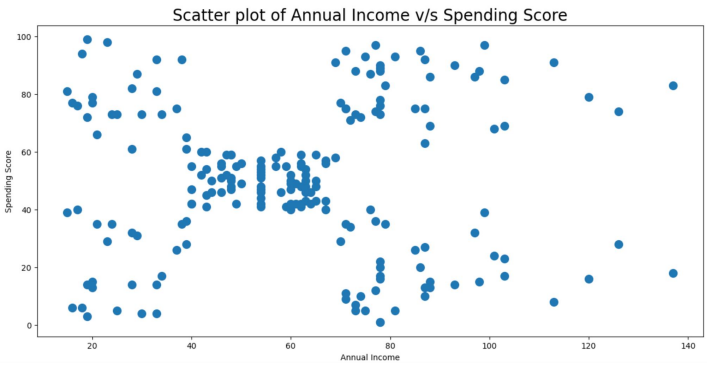


Figure2 Scatter plot of Annual Income and Spending Score

2. Data Collection and Data Description

This project uses the Mall Customer Segmentation dataset from Kaggle, which includes 200 customers and features such as Gender, Age, Annual Income, and Spending Score. There are no missing values, and CustomerID is excluded from the modeling process.

For the analysis, Age, Annual Income, and Spending Score are selected to represent customer demographics and purchasing behavior. Since these variables are on different scales (e.g., age versus income), the data is standardized prior to clustering. This ensures that each feature contributes equally to the analysis, allowing for a fair and balanced grouping of customers.

Table1 Descriptive statistics of the Mall Customer Segmentation dataset

	CustomerID	Age	Annual Income(k$)	Spending Score(1-100)
count	200.000000	200.000000	200.000000	200.000000
mean	100.500000	38.850000	60.560000	50.200000
std	57.879185	13.969007	26.264721	25.823522
min	1.000000	18.000000	15.000000	1.000000
25%	50.750000	28.750000	41.500000	34.750000
50%	100.500000	36.000000	61.500000	50.000000
75%	150.250000	49.000000	78.000000	73.000000
max	200.000000	70.000000	137.000000	99.000000

3. Clustering Models

3.1 K-Means Clustering

3.1.1 Model Description

K-Means clustering was used as the baseline model for customer segmentation. It is a distance-based unsupervised learning method that partitions observations into a predefined number of clusters by minimizing within-cluster variation. In this analysis, K-Means was applied to three standardized numerical variables: Age, Annual Income, and Spending Score. Standardization was necessary because these variables are measured on different scales; without scaling, variables with larger numerical ranges could dominate the distance calculations.

3.1.2 Model Establishment and Results

To determine the optimal number of clusters, values of k from 2 to 10 were evaluated using both inertia and silhouette score. Inertia decreases monotonically as k increases, and is therefore primarily used to identify the point at which additional clusters provide diminishing returns. In contrast, the silhouette score provides a more informative measure of clustering quality, as it captures both within-cluster compactness and between-cluster separation.

The silhouette score increased from 0.3355 at k = 2 to a maximum value of 0.4274 at k = 6, after which no consistent improvement was observed(Figure3). Meanwhile, the elbow plot indicated that the reduction in inertia became more gradual beyond k = 5 or k = 6. Based on these observations, k = 6 was selected as the optimal number of clusters for the K-Means model.

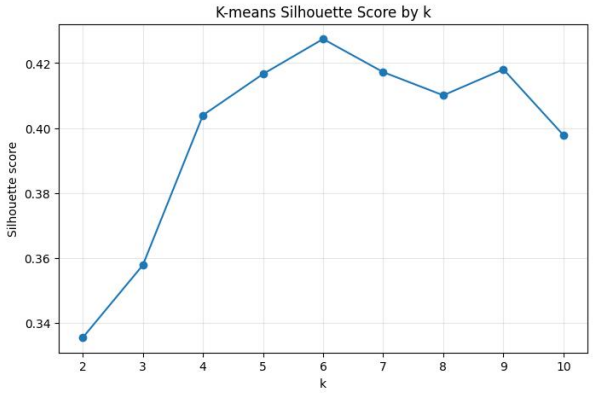


Figure 3. K-Means silhouette

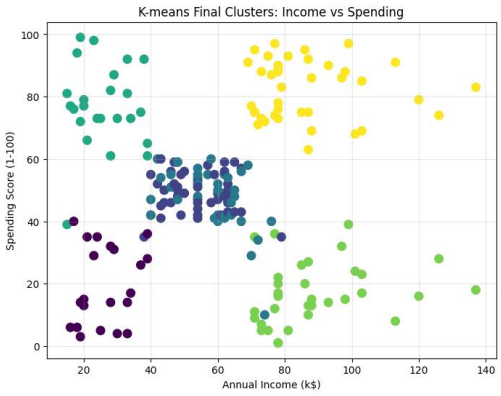


Figure 4. Final K-Means clusters

Using k = 6, the final K-Means model achieved an inertia of 133.8683 and a silhouette score of 0.4274. The six clusters were reasonably balanced in size, with group counts of 24, 21, 45, 38, 39, and 33 customers. The cluster profiles show meaningful differences across age, income, and spending behavior(Appendix2 Table1). For example, Group 5 represents relatively young, high-income, high-spending customers, with an average age of 32.69, average income of 86.54k, and average spending score of 82.13. In contrast, Group 6 represents high-income but low-spending customers, with an average income of 88.94k but an average spending score of only 16.97. The model also identifies lower-income high-spending customers, lower-income low-spending customers, and middle-income moderate-spending groups.

The final K-Means scatter plot using Annual Income and Spending Score provides a clear visual summary of the segmentation result(Figure4). Although the model was trained using all three numerical variables, the income-spending plot is useful for interpretation because these two variables show the most visible customer behavior pattern. The model separates high-income customers into high-spending and low-spending segments, which is important for marketing decisions. It also shows that customers with similar income levels can have very different spending scores, meaning income alone is not enough to explain customer behavior.

3.1.3 Model Validation

To check whether the clusters were mainly driven by gender, a chi-square test was conducted between gender and K-Means cluster assignment. The p-value was 0.5875, which is greater than 0.05. This suggests that there is no statistically significant relationship between gender and the K-Means clusters. In other words, the customer groups are mainly formed by numerical behavior and demographic variables such as age, income, and spending score, rather than by gender alone.

The stability test also supports the reliability of the K-Means result. Across 30 random seeds, the mean silhouette score was 0.4276 with a very small standard deviation of 0.0004. The average ARI across runs was 0.9966, with a minimum of 0.9772 and a maximum of 1.0000. This means that the final K-Means clustering result is highly stable and reproducible. Overall, K-Means provides a strong baseline model because it has good clustering quality, stable results, and clear business interpretation.

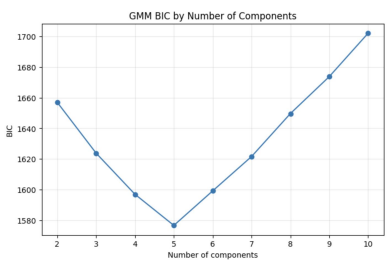
3.2 Gaussian Mixture Model (GMM)

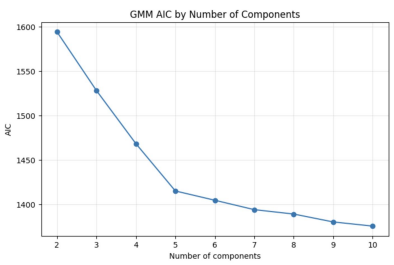
3.2.1 Model Description

Gaussian Mixture Model(GMM) is a probabilistic clustering method used to group customers based on similar characteristics. In this project, the model uses Age, Annual Income, and Spending Score to identify different customer groups. Unlike K-means, which assigns each customer to only one cluster, GMM uses soft clustering. This means each customer has a probability of belonging to different groups. Because of this, GMM is more flexible than K-means and can handle overlapping customer behavior better.

3.2.2 Model Establishment and Results

To choose the best number of components, the model tested values of k from 2 to 10. Three evaluation measures were used: BIC, AIC, and silhouette score. Lower BIC and AIC values indicate a better model fit, while a higher silhouette score indicates better clustering quality.
From the tuning table(Appendix2 Table2), the BIC value decreases until k=5 and then starts increasing again. The lowest BIC value is approximately 1576.71 at k=5. This suggests that 5 components provide the best balance between model fit and model complexity. The AIC value continues to decrease slightly after k=5, but the improvement becomes very small. At the same time, the silhouette score reaches its highest value around 0.4064 at k=5. Therefore, the project selects k=5 as the final number of components for the GMM model.





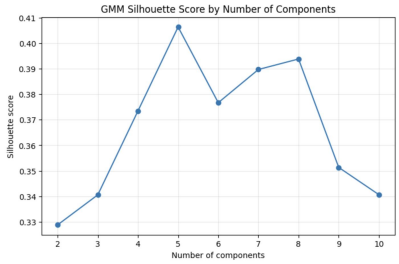


Figure5 GMM BIC Figure6 GMM AIC Figure7 GMM Silhouette

Through the evaluation using BIC and AIC indicators, BIC reached its minimum value at k=5 and then began to increase, indicating that 5 clustering components are the optimal point for balancing model complexity and fit (Figure5 and Figure6). Meanwhile, the Silhouette Score reached its highest value (approximately 0.4064) at k=5, further verifying that this number of clusters can provide the best classification quality(Figure7).

The final GMM model performed robustly(Appendix2 Table3), with an average maximum posterior probability of up to 0.9739, demonstrating that the model has strong confidence in classifying customers. The clustering results revealed significant group differences: there is a "core high-quality group" with high income and high consumption, a "potential mining group" with high income and low consumption, and a middle-income and middle-consumption group in the central area. This proves that relying solely on the income dimension cannot fully explain customer behavior.

In the two-dimensional distribution of "annual income" and "consumption score", GMM successfully identified 5 commercially significant market segments(Figure8). Compared to K-Means, GMM uses probability distribution modeling and generates smoother and more natural distribution-based clustering boundaries. This clustering method enables enterprises to formulate more precise differentiated marketing strategies for different groups with different risk preferences and consumption potential.

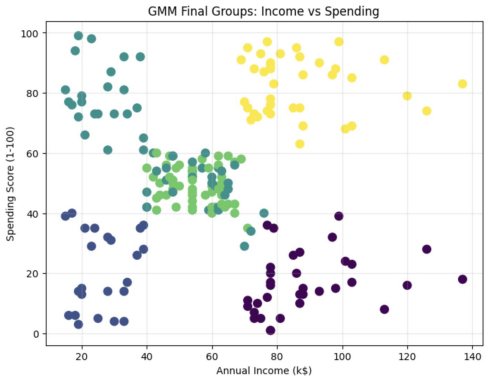


Figure8 Final GMM clusters

3.2.3 Model Validation

To test whether gender is related to the clustering result, a chi-square test was performed. The chi-square test gives a p-value of about 0.5408, which is larger than 0.05. This means there is no significant relationship between gender and the GMM clusters. Therefore, the clustering result is mainly driven by customer behavior and income instead of gender differences.

The stability of the GMM model was also tested using different random seeds. The stability table shows that the silhouette score and BIC values remain almost identical across different runs. The ARI is equal to 1.0, which means the clustering assignments are extremely stable and reproducible.

Overall, GMM provides a more flexible clustering approach than K-means because it allows soft clustering and probabilistic group assignment. However, in this dataset, K-means still achieves the best silhouette score at about 0.4274, while Spectral Clustering achieves about 0.4148. GMM achieves a silhouette score of about 0.4064, which is slightly lower than the other two methods. This suggests that although GMM is more flexible, the customer structure in this dataset is still captured more clearly by distance-based clustering methods such as K-means.

3.3 Spectral Clustering

3.3.1 Model Description

Spectral clustering is a graph-based clustering method that groups data points based on their similarity structure rather than relying on direct distance measures. Unlike K-means, which assumes spherical clusters and minimizes within-cluster variance, spectral clustering constructs a similarity graph and uses eigenvector decomposition of the Laplacian matrix to identify clusters in a lower-dimensional space. In this project, we use the *nearest neighbors* affinity to construct the similarity graph and apply K-means on the spectral embedding. This method is particularly suitable for capturing complex, non-linear cluster structures and provides more flexibility compared to traditional clustering approaches.

3.3.2 Model Establishment and Results

To determine the optimal number of clusters, we evaluated the silhouette score for different values of k ranging from 2 to 10(Figure9). The silhouette score increases significantly from k=2 and reaches its highest value at k=7 with a score of approximately 0.4148. After this point, the score slightly decreases or stabilizes, indicating that additional clusters do not improve clustering quality. Therefore, k=7 is selected as the optimal number of clusters for the spectral clustering model.

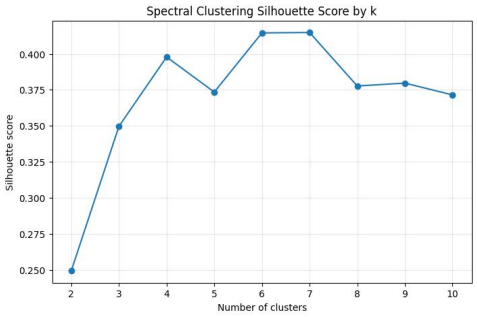


Figure9 Spectral Clustering Silhouette Score by k

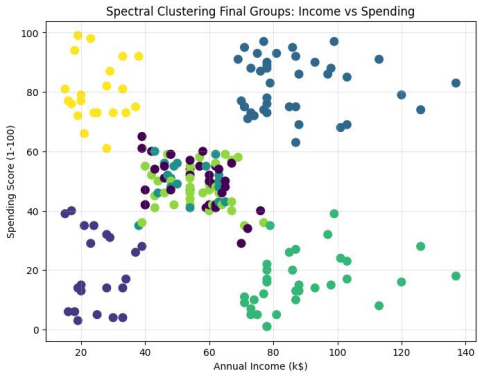


Figure10 Final Spectral Clustering plot

The final spectral clustering model uses k=7 and achieves a silhouette score of 0.4148. The cluster size distribution(Appendix2 Table4) shows that observations are relatively balanced across the seven clusters. The cluster profile table reveals distinct differences in age, income, and spending patterns among groups.

The scatter plot of Annual Income versus Spending Score further illustrates the segmentation structure(Figure10). Customers are divided into several meaningful groups, including high-income high-spending segments and high-income low-spending segments. Similarly, low-income customers are also separated into different behavioral groups. These results suggest that income alone cannot fully explain customer behavior, and spending patterns play a crucial role in defining customer segments. Overall, spectral clustering provides a flexible and effective segmentation that captures underlying non-linear structures in the data.

3.2.3 Model Validation

To examine whether the clustering results are associated with gender, a chi-square test was conducted using the contingency table of gender distribution across clusters. The chi-square statistic is 5.3422 with a p-value of 0.5007. Since the p-value is greater than 0.05, there is no statistically significant relationship between gender and cluster assignments. This suggests that the clustering results are not driven by gender, but rather by numerical features such as income and spending behavior.

Table2 Stability Summary Table

metric_mean	metric_sd	ari_mean	ari_min	ari_max
0.414818	5.64013e-17	1.0	1.0	1.0

To evaluate the robustness of the model, the spectral clustering algorithm was run multiple times with different random seeds. The stability results show that the clustering structure remains consistent across runs. The adjusted Rand index (ARI) is close to 1.0 across all runs, indicating

highly consistent cluster assignments(Table2). Additionally, the silhouette scores remain stable, confirming that the model produces reliable and reproducible clustering results.

3.4 Hierarchical Clustering

3.4.1 Model Description

Hierarchical clustering was used as an additional method to compare with K-means, GMM, and Spectral Clustering. Unlike K-means, which directly assigns observations into a fixed number of clusters, hierarchical clustering builds a tree-like structure showing how customers are gradually combined into larger groups. In this project, we used agglomerative hierarchical clustering with Euclidean distance and Ward linkage. Euclidean distance measures the distance between customers based on Age, Annual Income, and Spending Score, while Ward linkage helps form compact clusters by minimizing within-cluster variance. Therefore, this method is suitable for customer segmentation because it provides interpretable customer groups and a dendrogram that visually explains how the groups are formed.

3.4.2 Model Establishment and Results

The dendrogram shows how individual customers are gradually merged into larger clusters(Figure11). The vertical distance represents how different the merged groups are, so larger jumps indicate that the model is combining more distinct customer segments. The red horizontal line provides a visual reference for cutting the tree into several meaningful groups. Overall, the dendrogram suggests that the customer data has a clear hierarchical structure, but the final number of clusters still needs to be confirmed using a quantitative measure such as the silhouette score.

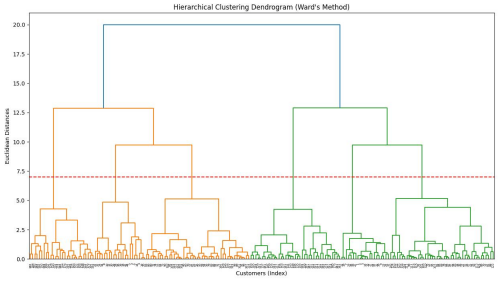


Figure11 Dendrogram

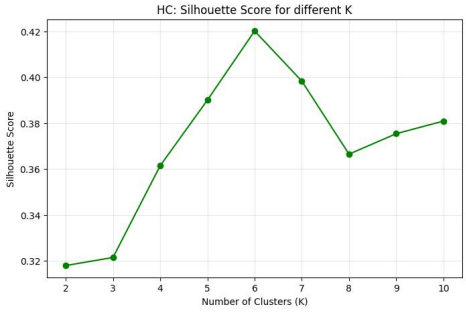


Figure12 Silhouette Score for Different K

The silhouette score plot was used to select the final number of clusters(Figure12). The score increases from about 0.318 at k = 2 to its highest value of 0.4201 at k = 6, showing that clustering quality improves as the model separates customers into more detailed groups. After k = 6, the score decreases, which suggests that adding more clusters may split meaningful groups into smaller and less useful subgroups. Therefore, k = 6 was selected as the final number of clusters for the hierarchical clustering model.

3.4.3 Model Validation

For external validation, a chi-square test was used to examine whether gender is related to the hierarchical cluster assignments. The result shows a chi-square statistic of 1.2871 and a p-value of 0.936258. Since the p-value is much larger than 0.05, there is no statistically significant relationship between gender and the hierarchical clusters. This suggests that the model is not simply separating customers by gender. Instead, the clusters are mainly formed based on numerical customer characteristics such as Age, Annual Income, and Spending Score.

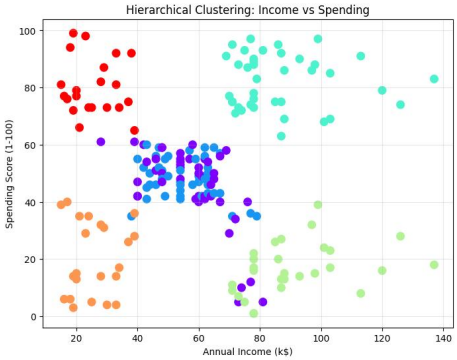


Figure13 Hierarchical Clustering

The final hierarchical clustering model uses k = 6 and produces six customer groups(Figure13). The scatter plot shows that income and spending score create several meaningful customer segments. Low-income customers are divided into high-spending and low-spending groups, which suggests that income alone cannot explain customer behavior. High-income customers are also separated into high-spending and low-spending groups, which is useful for business interpretation. High-income and high-spending customers may represent the most valuable segment, while high-income but low-spending customers may have strong purchasing power but lower current engagement. Overall, hierarchical clustering provides a stable and interpretable segmentation result.

4. Model Comparison

After testing K-Means, Hierarchical Clustering, GMM, and Spectral Clustering, we compared them all using k=6 groups to make the comparison fair. We focused on two main things: the overall quality of the clusters and how much the different models agreed with each other. This helps us decide which model is the most reliable for grouping customers.

To measure quality, we used the silhouette score, which shows how well each customer fits into their group. As shown in Figure 14, K-Means achieved the highest score, meaning its groups were the most organized and clearly separated. Hierarchical Clustering performed almost as well,

while GMM had the lowest score. This suggests that distance-based methods like K-Means are better at finding the geometric patterns in our customer data.

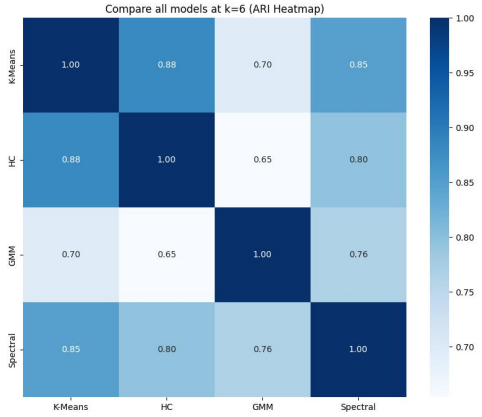


Figure14 Silhouette core comparison at k = 6

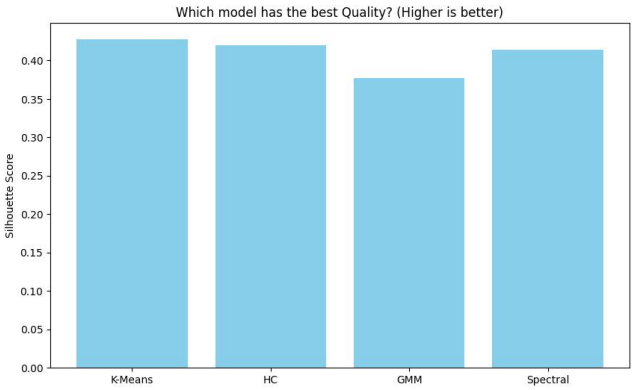


Figure15 ARI heatmap at k = 6

We also used the Adjusted Rand Index (ARI) to see if the models assigned customers to similar groups. As shown in Figure 15, K-Means and Hierarchical Clustering had a very high agreement of 0.88, and K-Means also agreed strongly with Spectral Clustering at 0.85. However, the ARI heatmap showed that GMM was less consistent with the others, proving that it was capturing a different and slightly weaker structure.

In conclusion, we chose K-Means as the best model for this project. It has the highest clustering quality and is supported by the other methods through high ARI scores. K-Means is also very easy to explain for business use because it provides clear centers for each group, such as "high-income high-spenders." While the other models were useful for comparison, K-Means offers the best balance of accuracy and clarity.

Appendix 1 Literature Review

Customer segmentation is a very important tool in modern marketing. It helps businesses divide a large group of customers into smaller groups based on their similarities, such as how much they spend or what they like to buy[2]. According to researchers, using machine learning to group customers allows companies to find "high-value" groups that they should focus on. Instead of looking at every person individually, businesses can create special marketing plans for different groups to save money and work more efficiently[1].

The most popular method for this task is the K-means algorithm. It is widely used because it is simple to understand and works very fast on large amounts of data[5]. However, K-means has a weakness: if the starting points (centroids) are chosen randomly, the results might not be accurate. To fix this, a better version called K-means++ was created[3]. This method chooses the starting points more carefully, which makes the clustering results more stable and reliable[3].

In practical studies, researchers often use a technique called the Elbow Method to decide how many groups to create. For example, in studies of retail shops and malls, researchers analyzed data like age, annual income, and spending scores[1]. By using K-means, they successfully sorted customers into five categories, such as "Careful" customers or "Target" customers[4]. To make sure the results are good, they also use scores like the Silhouette Score to check the quality of the groups[4].

Overall, these papers show that clustering is a powerful way to understand consumer behavior. While the basic K-means algorithm has some limits, newer versions and careful data preparation can solve these problems[5]. This project will use these methods to study mall customer data and find meaningful patterns that can help businesses make better decisions.

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Appendix 2 Visualization

Table1 Final K-Means Cluster Results

Kmeans_group	Age_mean	Age_count	Annual_Income_mean	Annual_Income_count	Spending_Score_mean	Spending_Score_count
Group 1	25.25	24	25.833333	24	76.916667	24
Group 2	45.52381	21	26.285714	21	19.380952	21
Group 3	56.333333	45	54.266667	45	49.066667	45
Group 4	26.684211	38	57.578947	38	47.789474	38
Group 5	32.692308	39	86.538462	39	82.128205	39
Group 6	41.939394	33	88.939394	33	16.969697	33

Table 2 GMM BIC, AIC, and Silhouette Score

k	BIC	AIC	silhouette_score
2	1656.962213	1594.294183	0.328788
3	1623.824883	1528.173679	0.340558
4	1596.999117	1468.36474	0.373447
5	1576.707936	1415.090385	0.406367
6	1599.229239	1404.628514	0.376684
7	1621.622622	1394.038724	0.389701
8	1649.655218	1389.088146	0.393814
9	1673.751039	1380.200794	0.35131
10	1702.049263	1375.515844	0.340630

Table3 Final GMM Cluster Results

GMM_group	Age_mean	Age_count	Annual_Income_mean	Annual_Income_count	Spending_Score_mean	Spending_Score_count
Group 1	45.217391	23	26.304348	23	20.913043	23
Group 2	24.830189	53	43.226415	53	62.037736	53
Group 3	53.392157	51	55.372549	51	48.980392	51
Group 4	32.692308	39	86.538462	39	82.128205	39
Group 5	41.647059	34	88.735294	34	16.764706	34

Table4 Final Spectral Cluster Results

Spectral_group	Age_mean	Age_count	Annual_Income_mean	Annual_Income_count	Spending_Score_mean	Spending_Score_count
Group 1	25.333333	23	25.095238	23	80.047619	23
Group 2	44.142857	45	25.142857	45	19.52381	45
Group 3	65.263158	66	53.105263	66	49.789474	66
Group 4	24.727273	24	55.272727	24	49.969697	24
Group 5	47.411765	48	56.147059	48	47.558824	48
Group 6	32.692308	32	86.538462	32	82.128205	32
Group 7	41.454545	42	89.090909	42	16.181818	42